Minh-Nhat Nguyen

(+84) 898-203-321 | 2512021580@student.hcmus.edu.vn | github.com/xdnhatnguyen | xdnhatnguyen.github.io

Professional Summary

A second-year Computer Science undergraduate with a solid mathematical foundation (**Calculus, Linear Algebra, Stochastic Systems**) and hands-on research in **Computer Vision** and **Generative Models (Diffusion Models, Flow Matching)**. Experienced in developing training-free generative methods and multimodal video retrieval pipelines. Seeking an **AI Engineer / Researcher** position in corporate **R&D** to advance applied computer vision and generative AI systems.

Education

University of Science, VNUHCM — Ho Chi Minh City, Vietnam
Bachelor of Information Technology – ***GPA: 3.74/4.0 (8.75/10.0)*** — *Aug. 2025 – 2029 (expected)*
Intended Major: Computer Science
Relevant Coursework (All 4.0/4.0): Calculus I & II, Linear Algebra, Data Structures & Algorithms, Programming Fundamentals
Advanced Online Coursework: **MIT 6.S184:** Generative AI with Stochastic Differential Equations

Le Quy Don Gifted High School — Da Nang, Vietnam
High School Diploma – Mathematics Specialization — *Aug. 2022 – May 2025*

Experience & Research

Undergraduate AI Researcher — March 2026 – Present
University of Science, VNUHCM — *Ho Chi Minh City, Vietnam*

- Developing a Training-Free method for Style Transfer using Latent Diffusion Models and Flow Matching, introducing latent patch-shuffling to eliminate content leakage and pixel-space grid artifacts.
- Optimizing generative algorithms by manipulating sampling trajectories, guidance terms, and cross-attention maps to control stylistic and structural outputs.

Honors & Awards

Quarter-finalist (Top 36 / 200+) — May 2026
Coding Challenge 2026 – Faculty of Information Technology, HCMUS — *Ho Chi Minh City, Vietnam*

Projects

Video Retrieval Pipeline – AI Challenge HCMC 2026 | *LanceDB, SigLIP, DP* — Aug. 2026

- Leading a team in the Ho Chi Minh City AI Challenge 2026 to build an end-to-end multimodal video retrieval pipeline addressing Known-Item Search (KIS), Visual Question Answering (VQA), and Temporal Alignment (TRAKE).
- Engineered a high-throughput hybrid retrieval engine combining dense semantic embeddings (CLIP/SigLIP) with full-text search (BM25 via LanceDB) and Dynamic Programming (DANTE algorithm) for temporal sequence matching.
- Optimized candidate frame ranking, automated query parsing, and sub-second retrieval latency on large-scale video keyframe datasets.

Latent Patch-Shuffle Style Transfer | *PyTorch, Stable Cascade Diffusion* | Article Link — Aug. 2026

- Researched and implemented a training-free style transfer framework on latent diffusion models, utilizing latent patch-shuffling to retain core object geometry while preventing content leakage.
- Eliminated pixel-space grid line artifacts by shifting patch permutations into the latent manifold, achieving vivid artistic style adaptation across diverse domains (Cyberpunk, Starry Night, Poster).

Handwritten Digits Classification | *PyTorch, CNN, MLP* | GitHub — Nov. 2025

- Led a research team to build and train Convolutional Neural Network (CNN) and Multilayer Perceptron (MLP) architectures from scratch for the MNIST dataset.
- Emphasized rigorous evaluation by benchmarking and comparing the efficiency, loss convergence, and accuracy between the two architectures.

- Co-authored an article detailing mathematical derivations and comparative analysis (Article Link).

Gomoku (Tic-Tac-Toe) AI Bot | *C++, Algorithms* | GitHub — Dec. 2025

- Developed a classical AI opponent for the game of Gomoku using procedural programming in C++.
- Implemented the Minimax algorithm combined with Alpha-Beta pruning to optimize decision-making processes and evaluate game states efficiently.

Technical Skills

Core Knowledge: Deep Learning, Computer Vision, Generative Models (Flow Matching, Diffusion Models, SDEs), Multimodal Information Retrieval (LanceDB, FAISS, Vector Search, BM25)
Programming & Tools: Python, C/C++, PyTorch, LanceDB, HuggingFace (Transformers/CLIP), Git, LaTeX
Mathematics: Linear Algebra, Multivariable Calculus, Optimization, Stochastic Differential Equations (SDEs)